

EXHIBIT 23 09 95A - Pre Functional Checklist Sample
Cal Poly - Vista Grande Dining Facility

System Type: Air Handler Unit

Equipment Tag (ID): (AH-)

PFC FORM ID: ____

A. Submittal / Approvals

Submittal. The above equipment and systems integral to them are complete and ready for functional testing. The checklist items are complete and have been checked off only by parties having direct knowledge of the event, as marked below, respective to each responsible contractor. This prefunctional checklist is submitted for approval, subject to an attached list of outstanding items yet to be completed. A Statement of Correction will be submitted upon completion of any outstanding areas. None of the outstanding items preclude safe and reliable functional tests being performed.

<u>General Contractor</u>	<u>Date</u>
<u>Mechanical Contractor</u>	<u>Date</u>
<u>Electrical Contractor</u>	<u>Date</u>
<u>Controls Contractor</u>	<u>Date</u>

Pre-functional checklist items are to be completed as part of start-up & initial checkout, preparatory to functional testing.

- This checklist does not take the place of the manufacturer's recommended checkout and start-up procedures or report
- Items that do not apply shall be noted with the reasons on this form (N/A = not applicable, BO = by others)
- If this form is not used for documenting, one of similar rigor shall be used.
- General contractor shall assign checklist sections to the respective sub-contractors. General contractor shall be responsible to see that the checklist items by their sub-contractors are completed and checked off.

Approvals. This filled-out checklist has been reviewed. Its completion is approved with the exceptions noted in the issues log.

<u>Commissioning Authority</u>	<u>Date</u>
--------------------------------	-------------

B. Model Verification

Enter nameplate information; Verify installed units match approved submittals

	AH-1
Manufacturer	
Model	
Serial Number	
CFM	
Cooling Capacity (MBH)	
Heating Capacity (MBH)	

C. Pre-Functional Checks

Check (✓) if Okay. Enter "N/A" if not applicable. Enter note number if deficient.

Tag	Inspection Item	Equipment Tag		Comments
		AH-1		
1	DOCUMENTATION			
	<i>Verify if the following items have been submitted:</i>			
1.01	Manufacturer's cut sheets including performance data (if available)			
1.02	Installation and startup manual / plan			
1.03	O&M manuals			
1.04	Control sequences			

VISTA GRANDE FACILITY REPLACEMENT
MAJ 15-MJ0061

Tag	Inspection Item	AH-1		Comments
2	CABINET AND GENERAL INSTALLATION			
2.01	Permanent labels affixed - indoor and outdoor units			
2.02	Casing condition good: no dents, leaks, door gaskets installed			
2.03	Access doors close tightly - no leaks			
2.04	Boot between duct and unit tight and in good condition			
2.05	Vibration isolation equipment installed & released from shipping locks			
2.06	Maintenance access acceptable for unit and components			
2.07	Sound attenuation installed (if applicable)			
2.08	Thermal insulation properly installed according to specifications			
2.09	Instrumentation installed according to specifications (temperature gauges, pressure gages, flow meters, etc.)			
2.10	All the duct connections to the cabinet are air tight and insulated			
2.11	Filters installed and replacement type and efficiency label permanently affixed to housing - construction filters removed			
3	VALVES, PIPING AND COILS			
3.01	Pipe fittings complete and pipes properly supported			
3.02	Piping properly labeled			
3.03	Piping properly insulated			
3.04	Piping properly flushed			
3.05	No leaks apparent around fittings			
3.06	All coils are clean and are in good condition			
3.07	All condensate drain pans clean and slope to drain, per spec			
3.08	Valves installed in proper direction			
3.09	Valves properly labeled			
3.10	Pressure / Temperature (P/T) plugs and isolation valves installed (per drawings)			
3.11	Piping through exterior walls is weather sealed			
4	FANS AND DAMPERS			
4.01	Supply fan and motor alignment appear correct			
4.02	Supply fan belt tension and condition appear OK (if applicable)			
4.03	Supply fan protective shrouds for belts in place and secure (if applicable)			
4.04	Supply fan area clean			
4.05	Supply fan and motor properly lubricated			
4.06	Smoke and fire dampers installed properly per contract documents (proper location, access doors, appropriate ratings verified)			
4.07	All dampers close tightly			
4.08	All damper linkages have minimum play			
4.09	Filters clean and tight fitting			
4.10	Filter pressure differential measuring device installed and functional (magnahelic, inclined manometer, etc.)			
5	DUCTING			
5.01	Duct joint sealant properly installed			
5.02	No apparent severe duct restrictions			
5.03	Turning vanes installed in square elbows (if applicable)			
5.04	OSA intakes located away from pollutant sources & exhaust outlets			
5.05	Pressure leakage tests completed (if applicable)			
5.06	Branch duct control dampers operable (if applicable)			
5.07	Ducts cleaned as per specifications			
5.08	Balancing dampers installed as per drawings and TAB company site visit			
5.09	Sound attenuators installed (if applicable)			
6	ELECTRICAL			
6.01	Power disconnects in place and labeled			
6.02	All electric connections tight			
6.03	Proper grounding installed for unit and components			
6.04	Safeties in place and operable			
6.05	Breaker size and type correct			
6.06	Power disconnect located within sight of unit controller (per electrical code)			

VISTA GRANDE FACILITY REPLACEMENT
MAJ 15-MJ0061

Tag	Inspection Item	AH-1		Comments
7	CONTROLS			
7.01	Control system interlocks hooked up and functional			
7.02	Related thermostats are installed			
7.03	Related building automation system points are installed			
7.04	All control devices and wiring complete			
7.05	Sensors calibrated (per specifications)			
7.06	Specified sequences of operation and operating schedules have been implemented with all variations documented			
7.07	Specified point-to-point checks have been completed and documentation record submitted for this system			
7.08	Variable Frequency Drive (VFD): Related Pre-Functional Checklist (PFC) form completed			
8	VARIABLE FREQUENCY DRIVE (VFD)			
8.01	VFD powered (wired to controlled equipment)			
8.02	VFD interlocked to control system (checked by installer)			
8.03	Controlling sensor properly located & per drawings & calibrated			
8.04	Drive location not subject to excessive temperature (high/low), moisture, or dirt			
8.05	Drive size matches motor size			
8.06	Motor is rated for use with VFDs			
8.07	Internal setting designating the model is correct			
8.08	Input of motor FLA represents 100% to 105% of motor FLA rating			
8.09	Appropriate Volts vs Hz curve is being used			
8.10	Accel and decel times are around 10-50 seconds, except for special applications. Actual accel = _____ Actual decel = _____			
8.11	Upper frequency limit set at 100%, unless explained otherwise			
8.12	Lower frequency limit at 10-15% for VAV fans and around 10-30% for chilled water pumps; or as dictated in the specifications. Record actual for each unit.			
8.13	Motor full load speed setting. Acceptance: Equal to motor rating.			
8.14	Motor frequency setting. Acceptance: same as rated motor freq.			
8.15	Motor line voltage setting. Acceptance: same as rated motor voltage, usually 460 volts.			
8.16	No disconnects installed between VFD & motor without shutdown interlock to VFD			
8.17	Shutdown interlocks between VFD & motor verified to be operational			
8.18	Separate conduit for VFD incoming power & outgoing motor leads			
8.19	Bypass contactor operation verified; Fan rotation correct with bypass operation			
9	OPERATIONAL			
9.01	Supply fan rotation correct			
9.02	Supply fan - No unusual vibration or noise			
9.03	CHW & HHW valves stroke fully and easily and spanning is calibrated			
9.04	Valves verified to not be leaking through coils when closed at normal operating pressure			
9.05	The HOA switch properly activates and deactivates the unit			
9.06	If supply fan is > 5 HP, measure line-to-line voltage imbalance. (If VFD-equipped, measure upstream of VFD). (% Imbalance = $100 \times (\text{max. deviation from avg.}) / \text{avg.}$) Fan motor (line-to-line voltage): a) Ph 1: _____, b) Ph 2: _____, c) Ph 3: _____ d) Average voltage: _____ e) % Imbalance: _____ f) Is imbalance less than 2%? (Yes/No)	a) _____ b) _____ c) _____ d) _____ e) _____ f) _____		
9.07	If supply fan is > 5 HP - Verify full load running amps. (If VFD-equipped, measure upstream of VFD). a) Rated Amps: _____ Rated FL amps x _____ svc factor = _____ (Max amps) b) Measured Amps: _____ c) Running less than max? (Yes/No)	a) _____ b) _____ c) _____		

Tag	Inspection Item	AH-1		Comments
10	FINAL			
10.01	If unit is started and will be running during construction: have quality filters on RA grills, etc. to minimize dirt in the ductwork and coils and in any finished areas.			
10.02	Safeties installed and safe operating ranges for this equipment provided to the commissioning agent			
10.03	Startup report completed with this checklist attached			
10.04	Clean up of equipment and surrounding areas completed			
10.05	Construction filters removed before occupancy. Replace with clean filters at start of occupancy (or per specifications)			

D. Notes